TETHER: Supplementary Material

Anonymous Authors

***Abstract*— This document reports measurements that did not fit the page limit. It adds no claim to the main paper. Two of its sections (Sec. III and Sec. VII) contain results that *qualify* the main claims rather than strengthen them, and they are included for that reason.**

I. Complete Matched-Baseline Closure

Table I lists all fifteen policies of the D2 closure. The main paper reports a representative subset. Every row shares the bounded state machine, the raw re-anchor schedule, the SEA-RAFT alignment, the support definition and the fusion form; only the blending weight differs, and all 225 reports (15 methods $\times$ 5 backbones $\times$ 3 sequences) come from a single run under one frozen protocol.

TABLE I
Full SCARED-C D2 matched temporal-policy closure. Raw EPE is 0.276202. Positive reduction denotes improvement.

Method	EPE	Reduction
TETHER $H=6$	**0.265893**	3.73%
TETHER $H=4$	0.265998	3.69%
TETHER $H=8$	0.266473	3.52%
TETHER $H=2$	0.267858	3.02%
TETHER $H=1$	0.269331	2.49%
Fixed $w=0.5$	0.274562	0.59%
EMA2	0.274562	0.59%
Fixed $w=0.3$	0.274852	0.49%
Fixed $w=0.2$	0.275229	0.35%
EMA3	0.275386	0.30%
Fixed $w=0.1$	0.275684	0.19%
Warped raw previous	0.276298	−0.03%
FB confidence weight	0.282182	−2.17%
Warped recurrent $w=1$	0.286035	−3.56%
Continuous recurrence	0.329140	−19.17%
Oracle raw/memory	0.241023	12.74%

a) An identity worth stating.: `EMA2` and `Fixed` $w=0.5$ are the same policy under two names, and `EMA3` is $w=2/3$: under a recurrent state a fixed convex weight *is* an exponential moving average. The baseline family is therefore five fixed weights $w \in \{0.1, 0.2, 0.3, 0.5, 2/3\}$ over the shared bounded state, not two independent families. We state this because two nominally different baselines producing an identical number otherwise reads as a bug.

The Horizon Sweep, With Its Risk Side

The main paper states that $H=4$ is a risk–gain operating point rather than the EPE minimum, and points here for the sweep. The closure table above gives the EPE column; this is the same sweep with the intervention statistics that motivated the choice.

TABLE II
Bounded horizon on D2. Lower EPE is better; higher B^+/H^+ and lower NewBad1 are better.

H	EPE	B^+/H^+	NewBad1 (%)
1	0.269331	1.82	0.0518
2	0.267858	**1.85**	0.0543
4	0.265998	**1.85**	0.0627
6	**0.265893**	1.74	0.0790
8	0.266473	1.60	0.1282

The two columns disagree, and that disagreement is the whole reason the operating point is not simply the EPE minimum. EPE bottoms out at $H=6$, by 0.04 points over $H=4$. Over the same step the beneficial-to-harmful magnitude ratio falls from 1.85 to 1.74 and then to 1.60, while newly introduced Bad1 pixels rise from 0.0627% to 0.0790% to 0.1282% — roughly doubling between $H=4$ and $H=8$. Letting the corrected state live longer buys a sliver of mean accuracy and pays for it in intervention risk.

$H=4$ was frozen on this trade before the held-out split was opened. We report it as a choice rather than as an optimum, because on the metric most papers would quote it is not one.

II. The Oracle Ceiling Is Not the One We Report Against

The oracle of the main paper selects, per pixel, the better of raw disparity and aligned memory. It is therefore the ceiling of $w \in \{0,1\}$, whereas TETHER acts on $w \in [0,1]$, whose ceiling is the continuous oracle $w^\star = \mathrm{clip}\big((g-d^{\mathrm{raw}})/(\widetilde{m}-d^{\mathrm{raw}}),0,1\big)$.

TABLE III
Binary versus continuous oracle on D2, per backbone.

Backbone	raw	TETHER	$\{0,1\}$	$[0,1]$
S^2M^2-S	.2949	.2858	.2624	.2599
RAFT-Stereo	.3022	.2920	.2675	.2645
Stereo Anywhere	.3189	.2974	.2736	.2708
CREStereo	.2943	.2838	.2597	.2568
Fast-FoundationStereo	.3037	.2939	.2702	.2674

The correction is real but small. Only 6.4% of D2 pixels have the ground truth bracketed by the two candidates, so the two ceilings nearly coincide and the recovered fraction falls from 33.8% to 31.4%. We report it because the published figure is measured against a ceiling the method cannot actually be limited by, not because it changes the conclusion.

III. Motion-Compensated Multi-Horizon Fidelity

$\mathrm{TEPE}_{\mathrm{corr}}$ and OPW require a correspondence field, supplied here from the same SEA-RAFT flows used for warping. Table III gives the relative change after refinement, on the 142-channel configuration; the family is a diagnostic the paper does not build on, and the shipped head is being measured on it separately.

TABLE IV

$\mathrm{TEPE}_{\mathrm{corr}}$ (top) and OPW (bottom), relative change after refinement. Negative is better.

Split	Backbone	k=1	k=2	k=4	k=8
$\mathrm{TEPE}_{\mathrm{corr}}$					
D2	S^2M^2-S	+24.5%	+40.1%	+12.9%	+19.7%
D2	RAFT-Stereo	+24.8%	+41.1%	+16.6%	+23.8%
D2	Stereo Anywhere	+14.5%	+31.1%	+8.5%	+16.4%
D2	CREStereo	+22.2%	+39.7%	+15.3%	+24.3%
D2	Fast-FoundationStereo	+35.9%	+52.3%	+21.1%	+27.8%
D7	S^2M^2-S	-23.7%	-21.0%	-17.4%	-12.7%
D7	RAFT-Stereo	-12.2%	-6.1%	-5.6%	-5.7%
D7	Stereo Anywhere	-10.3%	-4.8%	-5.3%	-5.7%
D7	CREStereo	-13.6%	-6.8%	-6.7%	-6.4%
D7	Fast-FoundationStereo	-10.9%	-4.9%	-5.6%	-6.2%
OPW					
D2	S^2M^2-S	-8.7%	-3.5%	-1.8%	-1.2%
D2	RAFT-Stereo	-8.5%	-3.4%	-1.7%	-1.1%
D2	Stereo Anywhere	-12.9%	-5.8%	-3.0%	-1.8%
D2	CREStereo	-9.5%	-3.8%	-1.8%	-1.2%
D2	Fast-FoundationStereo	-7.3%	-3.1%	-1.6%	-1.2%
D7	S^2M^2-S	-29.8%	-22.8%	-13.5%	-7.6%
D7	RAFT-Stereo	-18.5%	-10.2%	-5.1%	-3.4%
D7	Stereo Anywhere	-18.1%	-9.9%	-5.1%	-3.4%
D7	CREStereo	-20.3%	-11.1%	-5.9%	-3.8%
D7	Fast-FoundationStereo	-18.5%	-10.0%	-5.2%	-3.6%

On held-out D7 both improve in all $20/20$ backbone–horizon cases. On development D2 they disagree: OPW improves everywhere while $\mathrm{TEPE}_{\mathrm{corr}}$ degrades everywhere.

This rules out a convenient explanation. One might attribute a rising temporal-error metric to a grid metric penalising legitimate scene motion, but $\mathrm{TEPE}_{\mathrm{corr}}$ *is* motion compensated and still degrades on D2. The two ask different questions: OPW asks whether consecutive predictions agree with each other, $\mathrm{TEPE}_{\mathrm{corr}}$ whether the predicted temporal change matches the true one. On the split used to select the operating point, the module therefore becomes more self-consistent while tracking the true change less faithfully. We report this against our own interest.

IV. Forensics of the Support Violation

The main paper states that an unmasked output lets zero-padded out-of-bounds memory become an invalid prediction. The supporting measurement is below: every offending pixel has an aligned memory of *exactly* zero, and the offenders arrive in bursts no longer than the state lifetime, so bounded recurrence contains the damage it cannot prevent.

Because an invalid prediction on valid support is penalised rather than discarded, at 10^4 mm per pixel, a few hundred

TABLE V

Invalid refined pixels under the unmasked output, and the state age at which they occur.

	Vid10_Liver_Med	Vid11_Liver_High
Support pixels	38,828,160	38,439,360
Invalid (unmasked)	300	265
Invalid (support-confined)	**0**	**0**
Frames affected	10	3
Aligned memory, min/max	0.0 / 0.0	0.0 / 0.0
Non-finite memory	0	0
State age 1/2/3/4	10/103/135/52	0/182/72/11

pixels out of 38.8 million decide the external-domain metric-depth verdict.

V. Per-Recording DRENDS Transfer

TABLE VI

Zero-shot DRENDS, relative change after refinement, all metrics and all five recordings. Negative is better.

Backbone	Recording	EPE	Bad3	d.MAE	d.RMSE
RAFT-Stereo (seen)	Vid10	-6.0%	-11.9%	-4.0%	-3.5%
	Vid11	-3.9%	-27.0%	-2.8%	-1.8%
	Vid12	-9.1%	-23.3%	-7.3%	-6.4%
	Vid13	-4.5%	-14.1%	-3.3%	-3.1%
	Vid14	-4.7%	-27.2%	-3.7%	-3.0%
CREStereo (unseen)	Vid10	-5.3%	-9.4%	-3.6%	-3.7%
	Vid11	-4.2%	-27.2%	-3.0%	-2.1%
	Vid12	-7.2%	-20.4%	-5.6%	-4.9%
	Vid13	-5.5%	-10.9%	-3.8%	-3.5%
	Vid14	-5.9%	-27.5%	-4.5%	-4.2%
Fast-Found. (unseen)	Vid10	-5.1%	-7.7%	-3.2%	-3.2%
	Vid11	-3.6%	-24.3%	-2.7%	-1.9%
	Vid12	-6.0%	-11.1%	-5.0%	-4.0%
	Vid13	-4.3%	-10.3%	-3.1%	-2.9%
	Vid14	-4.5%	-25.6%	-3.6%	-2.9%

All 60 cells here improve, as do the other 40 the main paper aggregates over. The per-recording pattern is set by the recording and not by the estimator: on Vid10 the EPE change is -6.0%, -5.3% and -5.1% for the seen and the two unseen backbones. That alignment, rather than the pooled mean, is what supports backbone independence.

VI. Stage-Wise Runtime

NVIDIA H100 PCIe, PyTorch 2.5.1+cu121, 144×180 evidence grid, 200 timed iterations after 30 warmup iterations.

Two observations. The 155k-parameter head costs 1.00 ms, 3.5% of the overhead: "compact head" is true but not the operative cost, since 91% is the two frozen flow passes. And the reverse pass costs 13.09 ms while feeding only the forward–backward confidence cue, whose direct use as a fusion weight worsens EPE by 2.17% (Table I); removing it would cut overhead by 45%. Substituting a constant for the cue at inference costs this head nothing measurable, which is reported in the main paper; a head *trained* without the reverse pass is still the experiment we have not run. The

TABLE VII

PER-STAGE TEMPORAL OVERHEAD, BOTH CONFIGURATIONS, MEASURED IN THE SAME SESSION ON AN OTHERWISE IDLE DEVICE.

Stage	TETHER (ms)	p95	142 ch (ms)
SEA-RAFT forward	13.088	13.238	13.173
SEA-RAFT reverse	13.089	13.186	13.174
Alignment	1.049	1.067	1.058
Evidence tensor	0.674	0.686	5.131
Fusion head	**1.002**	1.012	1.014
Total overhead	**28.903**		33.549
Peak VRAM	**95.2 MB**		137.9 MB

evidence column is where the two configurations separate: 0.67 ms against 5.13, because TETHER builds no learned features.

VII. THREE-DIMENSIONAL SURFACE CONSISTENCY

This section is superseded and kept for the record. It was written when we believed SCARED-C carried no usable per-frame camera pose, so that global reconstruction drift could not be measured; that belief was wrong, and the world-frame measurement in the main paper — which does accumulate geometry in one frame using the corrected poses — is the one that answers the question this section could only approach. What follows instead back-projects each frame's disparity to a metric cloud and compares it against the previous frame's cloud pulled through the same SEA-RAFT correspondence: frame-to-frame surface stability, not reconstruction drift. The numbers are the 142-channel configuration's; we have not re-run them on the shipped head, because the measurement they belong to has been replaced rather than updated.

TABLE VIII

FRAME-TO-FRAME 3D SURFACE STABILITY ON D2, POOLED MEANS. GT IS THE FLOOR SET BY REAL SCENE AND CAMERA MOTION.

Backbone	raw	refined	GT
point-to-point distance [mm]			
$\mathrm{S^2M^2}$-S	0.5016	0.4978	0.4758
RAFT-Stereo	0.5085	0.5034	0.4758
Stereo Anywhere	0.5060	0.4996	0.4758
CREStereo	0.5065	0.5004	0.4758
Fast-FoundationStereo	0.4890	0.4902	0.4758
surface-normal angle [deg]			
$\mathrm{S^2M^2}$-S	4.4878	4.3241	5.8586
RAFT-Stereo	5.1443	4.7297	5.8586
Stereo Anywhere	5.4499	4.8516	5.8586
CREStereo	5.6006	4.9958	5.8586
Fast-FoundationStereo	3.7381	3.8742	5.8586

Excess over the GT floor falls on four backbones of five, for instance $0.0326 \rightarrow 0.0276$ mm on RAFT-Stereo, and rises slightly on Fast-FoundationStereo. The effect is real but of the order of 0.005 mm, which is why it never appeared in the paper even before the world-frame measurement replaced it.

The normal metric carries a caveat that removes its floor entirely: GT normals are *noisier* than every prediction ($5.86°$ against 3.74–$5.60°$), because the pseudo-ground-truth is sparse and central differences amplify it. Predictions smoother than the reference cannot be scored against that reference, so the normal column should be read as a relative comparison between raw and refined only.

Further assumptions a reader may wish to challenge: the principal point is taken at the image centre with $f_y = f_x$, since the SCARED-C manifest carries only f_x and the baseline; flow correspondence stands in for true scene-point correspondence, gated by forward–backward consistency at library defaults; and both clouds live in their own camera frames, so the measure mixes surface change with rigid camera motion, a component the GT floor absorbs but does not isolate.

VIII. LARGER TRAINING DOES NOT MOVE THE OPERATING POINT

An augmented-training run keeps the architecture and enlarges the data. Both arms here are the 142-channel configuration, so this compares two training regimes and not two heads; we have not repeated it on the shipped head.

TABLE IX

HELD-OUT D7, CANONICAL VERSUS AUGMENTED TRAINING.

Metric	Canonical	Augmented
Mean EPE	**.4693**	.4706
HUR	**.3965**	.4457
H^+	**.0148**	.0235
B^+	.0376	**.0450**
BUR	.4951	**.5060**
Degraded-frame fraction	**.2642**	.3341
Worst frame degradation	**.1063**	.3958

The augmented model performs more beneficial updates and substantially more harmful ones. Scale shifts intervention aggressiveness without improving held-out risk–gain balance.

IX. EXTERNAL STABILISER CONTEXT

BiDAStabilizer [1] is the closest published plug-in stabiliser that can actually be run. The main paper reports the pooled comparison; this section records how it was made fair and what it looks like sequence by sequence.

Our first attempt was not comparable at all, for two independent reasons: the published reproduction ran RAFT-Stereo *robust* while we ran RAFT-Stereo *middlebury*, and it was scored under a different support contract (raw EPE 0.3029 against 0.2762). Subtracting those numbers would have been meaningless. Both differences are removed by reading the stored NPZ boundary the reproduction already wrote — the exact RGB and the exact disparity BiDAStabilizer consumed — running our module on that same input, and scoring all three conditions against the same ground truth on one prediction-independent support: GT coverage $\cap$ raw validity, which neither method can influence. The one remaining difference is the one that matters and is reported

TABLE X

PER-SEQUENCE COMMON-SUPPORT COMPARISON, RELATIVE CHANGE AGAINST RAW (NEGATIVE IS BETTER). FRAMES AND SUPPORT ARE IDENTICAL WITHIN A ROW GROUP.

Sequence	Method	EPE	Bad1	Bad3
keyframe_2 (1033 fr.)	BiDAStabilizer	**−1.45**	−4.01	**−5.31**
	TETHER	−0.76	**−15.65**	−4.00
keyframe_3 (1102 fr.)	BiDAStabilizer	**−2.32**	−2.99	**−27.99**
	TETHER	−0.38	**−4.81**	−16.22
keyframe_4 (2114 fr.)	BiDAStabilizer	+0.63	−5.81	−6.82
	TETHER	**−6.58**	**−17.76**	**−17.56**
Pooled (18.0M px)	BiDAStabilizer	−0.25	−4.83	−8.32
	TETHER	**−4.60**	**−13.58**	**−16.81**

rather than removed: BiDAStabilizer is bidirectional and consumes future frames.

The pooled row favours TETHER on every metric, but *four* of the nine per-sequence cells do not, and the bidirectional method takes both EPE cells on the two shorter sequences. On `keyframe_3` it improves EPE by 2.32% against our 0.38%, six times as much. The whole of our pooled EPE margin comes from `keyframe_4`, which is also the longest sequence and carries half the pixels. We report this rather than pooling it away: with three sequences that is well within what the split can produce, and a method that sees the future should be expected to win somewhere. The claim the comparison supports is the pooled one — that a strictly causal module beats a bidirectional one on the same input — and not that it does so everywhere.

X. ACQUISITION AUDIT OF THE COMPARISON SET

The main paper states that no published method in our category can be run head-to-head. That is a strong claim, so we record what was attempted, on which date, and with what response. Everything below was checked on 16 August 2026 from the same host and the same session; the same `curl` reached `huggingface.co` and `archive.org` throughout, so a failure below is the resource's, not our network's.

TABLE XI

WHAT IS ACTUALLY OBTAINABLE. "CATEGORY" IS CAUSAL + BLACK-BOX + ATTACHED TO A FROZEN STEREO NETWORK, WHICH IS OUR SETTING.

Method	Venue	Code	Weights
StereoDiffusion [2]	MICCAI'24	none	none
Neural Disp. Refinement [3]	3DV'21	yes	deleted
Lai et al. [4]	ECCV'18	yes	host dead
TC-Stereo [5]	ECCV'24	yes	yes
BiDAStabilizer [1]	ECCV'24	yes	yes

a) StereoDiffusion.: The only published method in our exact category. Its repository has one commit, 7.43 KiB of git objects, and contains a `.gitignore`, a `LICENSE` and a twelve-byte `README.md`. `git ls-remote -heads` shows only `main`. There is no implementation to run.

b) Neural Disparity Refinement.: Same input contract as ours (RGB plus a black-box disparity), spatial only. The code clones and imports. Both released Google Drive identifiers return HTTP 404 from `drive.usercontent.google.com/download`, from `docs.google.com/uc`, *and* from the plain `drive.google.com/file/d/.../view` landing page. A file that exists but is access-restricted answers 403 or serves an interstitial; 404 on the landing page means the object is gone. Hugging Face model search returns nothing for "neural disparity refinement" or "disparity refinement"; the GitHub releases API for the official repository returns an empty list; Zenodo returns zero records; ModelScope has no such model.

c) Lai et al.: The natural task-agnostic baseline: model-agnostic and causal at inference. Its `pretrained_models/download_models.sh` points at `vllab.ucmerced.edu`, which answers 200 for the directory listing and 404 for every checkpoint. The Wayback CDX index holds the project page, its stylesheets and the 34 MB supplementary PDF, and zero captures of `ECCV18_blind_consistency.pth` or its optimiser file — web crawlers archive HTML and skip large binaries. No Hugging Face mirror and no GitHub release exists.

d) TC-Stereo.: Obtainable and loadable: we found a configuration that matches all three released checkpoints with zero missing and zero unexpected keys (16.7 M parameters). Its temporal branch warps the previous disparity by a rigid 6-DoF reprojection assembled from per-frame camera pose, intrinsics and baseline.

We initially recorded that SCARED-C could not supply those poses. That was wrong, and we correct it here: SCARED-C exists precisely because SCARED's kinematics-propagated poses were too inaccurate, and it ships COLMAP-re-estimated per-frame poses together with refined intrinsics (`frame_data.tar.gz`, `intrinsics_colmap.yaml`). The rigid-scene assumption is also reasonable on this data, which is ex-vivo: the tissue is static and the endoscope moves. TC-Stereo's temporal path is therefore runnable on SCARED-C, and the row below — with the branch disabled — understates it a second time.

What remains true, and is the point worth making, is a difference in what each method requires. TC-Stereo consumes per-frame 6-DoF pose, intrinsics and baseline; TETHER consumes RGB, disparity and validity. On deformable data the rigid reprojection also stops being well-posed, which is where our optical-flow alignment still applies. We report the single-frame row as an integrated-architecture reference, and note the temporal comparison as available future work rather than as impossible.

e) Deep Video Prior.: Excluded on definition rather than on availability: it performs per-video test-time training over an entire clip, so it is not causal under any configuration.

The practical consequence is that our comparison budget goes where a comparison is actually possible: BiDAStabilizer on a common support (Section IX), TC-Stereo as the integrated reference, and the matched-baseline closure of Section I, in which every degenerate temporal policy shares our alignment, our support contract and our evaluation code and differs only in the decision rule.

XI. Seed Study in Full

Pre-registered before any seed result existed. The seed is the only field that varies from the locked recipe; the training script is a sibling of `train.py` that reuses its guards and overrides only that field. The paired bootstrap resamples backbone–sequence cells, seeds are averaged inside each resample, and four things were prohibited in advance: selecting the best seed, re-tuning any threshold after seeing seed results, changing the checkpoint-selection rule, and using D7 for any decision other than the final reported numbers.

TABLE XII

Relative Bad1 reduction (%) per backbone over three seeds of the 142-*channel* configuration, with the paired bootstrap interval and the fraction of resamples in which refinement failed to help. This is the study that found the development instability; the shipped head's own three seeds are below.

Split	Backbone	s0	s1	s2	mean ± std	CI_{95}	p
D2	CREStereo	8.26	6.62	3.57	6.15 ± 2.38	$[0.69, 12.74]$	.000
D2	Fast-Found.	7.91	5.74	2.65	5.43 ± 2.64	$[0.63, 12.86]$	.000
D2	RAFT-Stereo	6.95	5.40	3.00	5.12 ± 1.99	$[-0.12, 12.43]$	.035
D2	S^2M^2-S	7.40	6.70	4.45	6.18 ± 1.54	$[-0.31, 12.32]$	.038
D2	StereoAnywhere	19.65	15.95	12.43	16.01 ± 3.61	$[13.89, 19.26]$	.000
D7	CREStereo	3.86	4.51	4.32	4.23 ± 0.33	$[2.68, 7.27]$	.000
D7	Fast-Found.	3.97	4.08	3.76	3.93 ± 0.16	$[1.60, 8.58]$	.000
D7	RAFT-Stereo	3.95	5.35	4.99	4.77 ± 0.73	$[3.27, 7.72]$	.000
D7	S^2M^2-S	5.66	6.25	6.05	5.99 ± 0.30	$[1.50, 13.17]$	.000
D7	StereoAnywhere	6.40	7.07	6.73	6.73 ± 0.33	$[5.66, 12.12]$	.000
Pooled, 35 sequences		5.79	6.00	5.28	5.69 ± 0.37	$[4.11, 8.02]$	.000

Two features of this table are worth stating plainly rather than leaving for a reader to find.

First, three D2 rows have intervals that touch or cross zero. That is a property of the split, not of the method: D2 contributes three sequences, and a three-sequence bootstrap is genuinely wide. We prefer reporting it to quietly resampling pixels, which would have produced narrow intervals with no statistical meaning. Held-out D7 contributes four sequences and every one of its rows is strictly positive at $p = .000$ on both metrics.

Second, the seed ordering differs between splits in a way that is not noise. Pooled per split, the D2 Bad1 reduction falls monotonically across seeds — 10.78, 8.67, 5.75 — while D7 gives 4.78, 5.47, 5.19, in which seed 0 is the *weakest*. Ten of ten per-backbone comparisons follow the same direction. Our checkpoint-selection rule picks best D2 fused EPE, so seed 0 is by construction the run that looked best on D2; the D2 margin therefore carries selection optimism and the held-out margin does not.

a) The shipped head does not do this.: The same study on the 38-channel head, same recipe, same seeds, same rule:

TABLE XIII

Relative Bad1 reduction (%) per backbone over three seeds of the shipped head.

Split	Backbone	s0	s1	s2	mean ± std
D2	CREStereo	10.61	10.25	10.30	10.39 ± 0.20
D2	Fast-Found.	9.95	9.25	9.53	9.58 ± 0.35
D2	RAFT-Stereo	10.71	8.68	10.03	9.81 ± 1.03
D2	S^2M^2-S	9.13	9.57	9.68	9.46 ± 0.29
D2	StereoAnywhere	20.51	20.71	21.69	20.97 ± 0.63
D7	CREStereo	4.74	4.75	5.15	4.88 ± 0.24
D7	Fast-Found.	4.48	4.44	4.57	4.50 ± 0.06
D7	RAFT-Stereo	5.47	5.46	5.84	5.59 ± 0.22
D7	S^2M^2-S	6.40	6.42	6.94	6.58 ± 0.31
D7	StereoAnywhere	7.31	7.53	7.86	7.56 ± 0.28
Pooled		12.84	12.38	12.98	12.73 ± 0.32

No monotone fall, no inversion, and the pooled D2 deviation is 0.32 points against the 2.53 above. The instability was the evidence space, not the selection rule — which is unchanged between the two tables.

XII. TC-Stereo as an Integrated Reference, and Why It Is Weak Evidence

TC-Stereo [5] is the one temporally-aware competitor whose weights we obtained. Section X explains why its temporal branch cannot run on SCARED-C: it warps the previous disparity by a rigid 6-DoF reprojection built from per-frame camera pose, which the dataset does not provide and which deformable tissue would invalidate anyway. We ran it with the temporal branch disabled — its single-frame stereo path — over the same stored boundary as the BiDAStabilizer comparison, so the frames and the ground truth are bit-identical across methods.

TABLE XIV

Same 4249 frames, same ground truth, same support, 18.0M pixels. TC-Stereo is SceneFlow-pretrained and evaluated zero-shot with its temporal branch disabled. Read the caveat below before using this row.

Method	EPE	Bad1 (%)	Bad3 (%)	RMSE
Raw RAFT-Stereo robust	0.3029	2.087	0.186	0.4484
BiDAStabilizer	0.3021	1.987	0.171	0.4403
TETHER	**0.2889**	**1.804**	**0.155**	**0.4234**
TC-Stereo (SceneFlow)	0.5242	9.051	1.680	1.4059

a) The caveat, which is large.: This row understates TC-Stereo and we do not claim otherwise. The stored boundary is 144×180 with a median ground-truth disparity of 8.04 px (range 5.57–15.53), whereas TC-Stereo was trained at full resolution on disparity ranges an order of magnitude larger. Asking a from-scratch cost-volume matcher to operate two octaves below its training regime is not a fair test of the architecture. A refiner is far less exposed to this: BiDAStabilizer and TETHER both consume the raw disparity and correct it, so the resolution of the boundary constrains them much less than it constrains a matcher built around a correlation pyramid.

We report the row anyway, with this caveat attached, because the alternative — running TC-Stereo at full resolution on frames the other methods never saw — would buy fairness to one method by destroying the property that makes the whole table meaningful. The number is evidence that swapping in a different integrated architecture zero-shot is not a drop-in substitute for refining a strong in-domain frozen model. It is *not* evidence that TETHER outperforms TC-Stereo, and we make no such claim.

b) One detail of the support.: We planned two supports, ground truth alone and ground truth intersected with raw validity. On this boundary they coincide exactly: `raw_valid` is true everywhere, so both conditions select the same 17,999,475 pixels. The intersection is reported as the definition because it is the one that would matter on a boundary where the backbone declines pixels; here it is vacuous.

XIII. Analysis, in Full

The main paper compresses this section to a summary. Nothing here is new evidence; it is the interpretation of results already reported, kept intact because the page limit is on the submission and not on the supplementary.

a) What Makes TETHER Work, and Why So Few Parameters Suffice: The closure isolates four requirements. Geometric alignment is necessary but not sufficient: the aligned memory carries a 12.74% oracle ceiling, yet using it indiscriminately fails, since warped raw memory yields no meaningful gain and both direct FB-confidence weighting and full recurrent replacement worsen EPE. Under identical evidence and support the learned policy obtains more than six times the gain of the best fixed blend, and the recurrent state must stay bounded because continuous propagation accumulates error and collapses. The mechanism is motion alignment plus learned graded intervention plus bounded recurrent state, not generic smoothing.

This also explains the parameter count. The head never solves stereo matching: it receives two already meaningful candidates, d_t^{raw} and $\widetilde{m}_t$, plus evidence describing their disagreement, motion reliability and validity, so its problem is closer to judging whether temporal evidence is useful than to predicting disparity from RGB. Convex fusion further restricts the action space to a position between the two candidate geometries. We previously offered that constraint as a plausible reason why one small head transfers to unseen architectures. The pre-registration required that claim to be withdrawn rather than softened if the relaxed-convexity ablation matched or beat the canonical model on the held-out endpoint. It did, by the largest margin in the table, so the claim is withdrawn: we do not know why the head transfers.

b) The Oracle Gap Is the Next Problem: TETHER captures 33.8% of the raw-versus-memory oracle gain and 31.4% of the continuous oracle its convex action space could reach; only 6.4% of D2 pixels have the ground truth bracketed by the two candidates, so the ceilings nearly coincide. Useful evidence already exists; the open problem is discriminating helpful from harmful intervention.

c) The Fusion Weight Is Not an Uncertainty: Recovering the effective weight exactly from the convex form, $w_t = (d_t^{\mathrm{fused}} - d_t^{\mathrm{raw}})/(\widetilde{m}_t - d_t^{\mathrm{raw}})$, and scoring it as a predictor of harmful intervention gives AUROC 0.393–0.408 on D2 — below chance on the split it was selected on — and 0.488–0.507 on held-out D7, i.e. chance; the raw–memory disagreement and the update magnitude behave the same way. The reliability curve inverts between splits — for RAFT-Stereo, across the populated weight bins, the harm rate *falls* with w_t on D2 ($0.396 \to 0.287$) and *rises* on D7 ($0.250 \to 0.362$) — so w_t is not a weak risk score but a sign-unstable one, and a threshold tuned on development would select the wrong pixels held out. It is an intervention magnitude, not an uncertainty, and closing the oracle gap needs evidence the head does not receive.

d) Reconciling this with the intervention table: Taken alone, the previous paragraph appears to contradict the intervention-quality table of the main paper: if w_t carries no information about harm on D7, how does the same policy reach B^+/H^+ between 2.18 and 3.64 there? The two statements are about different things, and the aggregate gain factorises exactly into them. Writing $\delta e = e^{\mathrm{fused}} - e^{\mathrm{raw}}$ per intervened pixel,

$$\frac{B^+}{H^+} = \underbrace{\frac{\#\{\delta e < 0\}}{\#\{\delta e > 0\}}}_{\text{count}} \times \underbrace{\frac{\mathbb{E}[|\delta e| \mid \delta e < 0]}{\mathbb{E}[|\delta e| \mid \delta e > 0]}}_{\text{magnitude}}. \tag{1}$$

Pooled over the five backbones this evaluates to $2.861 = 1.294 \times 2.212$ on held-out D7 and $2.028 = 1.114 \times 1.820$ on D2. On D7, 56.4% of interventions improve the pixel and the improvements are $2.2\times$ larger than the regressions.

Neither factor requires w_t to rank anything. The policy shifts the *distribution* of δe favourably, which is what that table measures; the AUROC asks the strictly harder question of whether the scalar the head emits identifies *which* pixels fall in the harmful 43.6%, and on held-out data it does not. A policy can be reliably good on average and completely uninformative per pixel: those are independent properties, and TETHER has the first without the second.

This sharpens rather than weakens the oracle-gap claim. The remaining 67% of the oracle is not reachable by thresholding anything the head already computes, because the quantity it computes is a magnitude and the gap is a ranking problem.

e) Evaluation Matters: The framework exposes behaviour a mean would hide: on DRENDS mean and percentile geometry improve while RMSE worsens; longer horizons slightly improve mean D2 EPE while degrading B^+/H^+, NewBad1 and the frame tails. Temporal refinement is better read as an *intervention* than as a final estimator.

One reported quantity failed an audit and is withdrawn: FrameDegradation's CatastrophicFraction is definitionally identical to PositiveFraction at the default threshold, so we report only the degraded-frame fraction $\Pr(D_t > 0)$. The supplementary gives the derivation.

XIV. Intervention Quality per Backbone

The main paper quotes the ranges; this is the table behind them. B^+/H^+ is the beneficial-to-harmful error-magnitude ratio and "Rec./New" the ratio of recovered to newly introduced Bad1 pixels, both on the prediction-independent protocol support.

TABLE XV
TETHER INTERVENTION QUALITY, SEED 0. ALL RATIOS EXCEED ONE; HIGHER IS BETTER. "REC./NEW" IS RECOVEREDBAD1/NEWBAD1.

Backbone	D2 B^+/H^+	D2 Rec./New	D7 B^+/H^+	D7 Rec./New
S^2M^2-S	1.39	3.8×	2.16	7.5×
RAFT-Stereo	1.58	4.9×	2.92	4.7×
Stereo Anywhere	3.07	9.9×	3.25	11.1×
CREStereo	1.70	5.0×	2.57	4.7×
Fast-FoundationStereo	1.57	5.3×	2.26	7.8×

XV. ABLATIONS IN FULL

The main paper reports held-out EPE and Bad1 for the four pre-registered ablations. This is the same evaluation on both splits and four metrics.

TABLE XVI
RELATIVE REDUCTION (%) AGAINST THE SAME RAW PREDICTIONS. THE RAW BASELINE IS IDENTICAL TO SIX DECIMALS ACROSS ALL SIX RUNS, SO THE ROWS ARE DIRECTLY COMPARABLE. CANONICAL, A2 AND A3 ARE MEANS OF THREE SEEDS; A1 AND A4 ARE SINGLE RUNS. "CANONICAL" IS THE 142-CHANNEL CONFIGURATION THE DEVIATIONS WERE REGISTERED AGAINST; A2 IS THE HEAD THE PAPER SHIPS.

	D2 development				D7 held out			
	EPE	Bad1	Bad3	RMSE	EPE	Bad1	Bad3	RMSE
Canonical	3.55	8.40	16.71	12.97	5.06	5.15	4.29	5.60
A1	4.18	8.81	15.76	13.11	4.93	4.88	4.12	5.46
A2	4.13	12.73	19.64	13.87	5.61	5.84	5.18	6.02
A3	3.85	8.10	18.34	13.10	5.24	5.68	5.53	5.89
A4	5.53	12.44	19.26	13.58	0.05	6.02	5.32	6.32

Three things are visible here that the main table has no room for.

First, A4's failure is specific rather than general. It is the best variant in the table on Bad1, Bad3 and RMSE on *both* splits, and its held-out EPE gain is 0.05%. A quadratic metric improving while a linear one collapses rules out occasional large errors — outliers hurt RMSE first — and leaves broad degradation of pixels that were already close, which the mean registers and thresholds and squared errors do not.

Second, A1 and A2 differ in a way that makes them one experiment rather than two. A1 removes the 64 raw appearance channels but keeps the appearance self- and cross-correlations computed from those same features; it is indistinguishable from canonical. A2 removes the correlations as well, and is better on all four held-out metrics. The more appearance-derived evidence leaves the interface, the better the head generalises — which is what one would expect of evidence that ties the decision to the training domain.

Third, D2 and D7 disagree about A1. On D2 it is close to canonical on Bad1 (8.81 against 8.40) but better on EPE (4.18 against 3.55); on D7 it is 0.27 points below on the registered endpoint, inside the 0.37-point threshold and so unreadable rather than worse. This is the same development-versus-held-out inversion the seed study found, and it is the reason the pre-registration named a held-out endpoint before any variant was trained.

a) What happened next.: A2 was the strongest variant and was not promoted on the strength of a held-out win, because choosing on the held-out split is what makes a held-out split stop being one. It was given the same three-seed treatment the canonical model has, and the comparison that could promote it was fixed on D2, the development split, before those seeds were trained. It cleared that bar by 4.33 points with a comparable spread, so it was promoted, and it is the head the main paper proposes. A3 was then given the same three-seed treatment for the opposite reason — it beat its base by more than the declared threshold on the registered endpoint — and it still does (5.68 ± 0.16 against 5.15 ± 0.34). We decline it on the $16\times$ context-branch compute rather than on accuracy, and say so in the main paper rather than presenting the rejection as an evidential one.

XVI. A METRIC FIELD WE DO NOT REPORT

FrameDegradation's CatastrophicFraction was numerically identical to PositiveFraction on every inspected row. The cause is definitional: the two are $\Pr(D_t > \tau_{\text{cat}})$ and $\Pr(D_t > 0)$, and the catastrophic threshold defaults to $\tau_{\text{cat}} = 0$, which the evaluators never override. At that setting they coincide by construction. We therefore report only the degraded-frame fraction, defined explicitly as $\Pr(D_t > 0)$.

REFERENCES

[1] J. Jing, Y. Mao, A. Qiu, and K. Mikolajczyk, "Match Stereo Videos via Bidirectional Alignment," in *Proc. European Conf. Computer Vision (ECCV)*, 2024.

[2] H. Xu, C. Xu, and S. Giannarou, "StereoDiffusion: Temporally consistent stereo depth estimation with diffusion models," in *Proc. MICCAI*, 2024.

[3] F. Aleotti *et al.*, "Neural disparity refinement for arbitrary resolution stereo," in *Proc. Int. Conf. 3D Vision (3DV)*, 2021.

[4] W.-S. Lai *et al.*, "Learning blind video temporal consistency," in *Proc. European Conf. Computer Vision (ECCV)*, 2018.

[5] Z. Zeng *et al.*, "Temporally consistent stereo matching," in *Proc. European Conf. Computer Vision (ECCV)*, 2024.

XVII. MATERIAL MOVED FROM THE MAIN PAPER

The main paper is eight pages. Four objects that were in it are now here in full, and nothing measured was deleted to make room.

a) Table: out-of-domain head-to-head with BiDAStabilizer.: Five DRENDS recordings, 117.4M pixels of common support, neither method in domain.

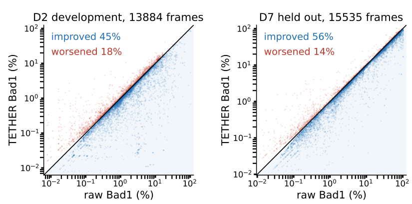
Method	Causal	EPE	Bad1 (%)	Bad3 (%)	RMSE
Raw	—	1.199	44.92	4.54	1.876
BiDAStabilizer	no	1.175	44.81	4.32	1.776
TETHER	**yes**	**1.131**	**43.39**	**3.94**	**1.710**

The two best columns belong to different rows. The bidirectional baseline is the more temporally stable method and the less accurate one, at every lag and in each of the five recordings.

b) Table: per-backbone DRENDS transfer.: Relative change in %, negative is better; each entry is the mean over the five recordings.

Backbone	Status	EPE	Bad3	Depth MAE	Depth RMSE
S^2M^2-S	seen	-4.86	-18.44	-3.65	-3.06
RAFT-Stereo	seen	-5.63	-20.71	-4.22	-3.58
Stereo Anywhere	seen	-6.21	-23.54	-4.60	-4.09
CREStereo	unseen	-5.62	-19.07	-4.12	-3.67
Fast-FoundationStereo	unseen	-4.70	-15.80	-3.50	-2.97
All 25 cells		-5.40	-19.51	-4.02	-3.47

c) Figure: per-frame Bad1 scatter.: Raw against refined for every scored frame of each split — five backbones, all sequences, no selection. On held-out D7, 56% of $15{,}535$ frames improve against 14% worsened; on development D2, 45% against 18%. The worsened band hugs the diagonal while the improved mass extends far from it, which is the count-versus-magnitude asymmetry behind B^+/H^+.



d) TC-Stereo, evidence-space construction, head widths, stage timing.: These already have sections of their own above.